1.62 Shortest Distances in 3D: Points and Lines

Worked Solutions

1)

The closest point on the x-axis to $(2, 5, -3)$ is $(2, 0, 0)$, so the distance is

$$\sqrt{(2-2)^2 + (5-0)^2 + (-3-0)^2} = \sqrt{34} \text{ units.}$$

Alternatively, let $P = (\lambda, 0, 0)$ be on the x-axis. At the closest point, $(P - A) \cdot (1, 0, 0) = 0$, so $\lambda - 2 = 0$ and $P = (2, 0, 0)$, giving the same distance. Using the vector product,

$$h = \frac{|(1,0,0) \times (2,5,-3)|}{|(1,0,0)|} = \frac{|(0,3,5)|}{1} = \sqrt{34} \text{ units.}$$

2)

The closest point on the z-axis to $(1, 8, -9)$ is $(0, 0, -9)$, so the distance is $\sqrt{1^2 + 8^2} = \sqrt{65}$ units. Alternatively, for $P = (0, 0, \lambda)$ on the axis, perpendicularity gives $(P - A) \cdot (0, 0, 1) = \lambda + 9 = 0$, so $P = (0, 0, -9)$. The vector-product method gives

$$h = \frac{|(0,0,1) \times (1,8,-9)|}{1} = \frac{|(-8,1,0)|}{1} = \sqrt{65} \text{ units.}$$

3)

The shortest distance is the perpendicular distance from O to the line through P perpendicular to $\overrightarrow{OP}$, so it is $|OP|$:

$$|OP| = \sqrt{6^2 + 12^2 + (-5)^2} = \sqrt{205} \text{ units.}$$

4)

A point on the line has position vector $(1 + 2\lambda, 3 + \lambda, 6 + 5\lambda)$. At the closest point to the origin, this vector is perpendicular to the line direction $(2, 1, 5)$:

$$2(1 + 2\lambda) + (3 + \lambda) + 5(6 + 5\lambda) = 0 \quad \Rightarrow \quad 30\lambda + 35 = 0, \quad \lambda = -\frac{7}{6}.$$

The closest point is $(-\frac{4}{3}, \frac{11}{6}, \frac{1}{6})$, so the distance is

$$\sqrt{\left(-\frac{4}{3}\right)^2 + \left(\frac{11}{6}\right)^2 + \left(\frac{1}{6}\right)^2} = \sqrt{\frac{31}{6}} \text{ units.}$$

Alternatively, using $\mathbf{d} = (2, 1, 5)$ and $\mathbf{p} - \mathbf{a} = (-1, -3, -6)$,

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{|(9, 7, -5)|}{\sqrt{30}} = \sqrt{\frac{31}{6}} \text{ units.}$$

5)

Let P lie on the line, so $\overrightarrow{OP} = (5-2\lambda, 3+\lambda, -4+3\lambda)$. With $A = (2,1,1)$, the closest point satisfies

$$(P-A)\cdot(-2,1,3) = (3-2\lambda, 2+\lambda, -5+3\lambda)\cdot(-2,1,3) = 0,$$

so $14\lambda - 19 = 0$ and $\lambda = \frac{19}{14}$. Thus $\overrightarrow{AP} = (\frac{2}{7}, \frac{47}{14}, -\frac{13}{14})$ and

$$AP = \sqrt{\left(\frac{2}{7}\right)^2 + \left(\frac{47}{14}\right)^2 + \left(-\frac{13}{14}\right)^2} = \frac{3\sqrt{266}}{14} \text{ units.}$$

Alternatively, with $\mathbf{d} = (-2,1,3)$ and $\mathbf{p}-\mathbf{a} = (-3,-2,5)$,

$$h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{|(11,1,7)|}{\sqrt{14}} = \frac{3\sqrt{266}}{14} \text{ units.}$$

6)

For a point $P = (1+\lambda, \lambda, 1)$ on the line, the vector from the centre $A = (-4,3,8)$ is $(5+\lambda, \lambda-3, -7)$. The shortest distance to the line occurs when this is perpendicular to the line direction $(1,1,0)$:

$$(5+\lambda)+(\lambda-3) = 0 \quad \Rightarrow \quad \lambda = -1.$$

Then $AP = (4,-4,-7)$, so $r = |AP| = \sqrt{4^2+(-4)^2+(-7)^2} = 9$. Alternatively,

$$r = \frac{|(1,1,0)\times((-4,3,8)-(1,0,1))|}{\sqrt{2}} = \frac{|(7,-7,8)|}{\sqrt{2}} = 9.$$

7)

The line is $\mathbf{r} = (1,0,-2)+\lambda(2,1,1)$. For $A = (1,0,1)$, the closest point satisfies

$$(2\lambda, \lambda, \lambda-3)\cdot(2,1,1) = 0 \quad \Rightarrow \quad 6\lambda - 3 = 0, \quad \lambda = \frac{1}{2}.$$

Thus the shortest distance is

$$\sqrt{1^2 + \left(\frac{1}{2}\right)^2 + \left(-\frac{5}{2}\right)^2} = \sqrt{\frac{15}{2}} = \frac{\sqrt{30}}{2} \text{ units.}$$

Alternatively, $\frac{|(2,1,1)\times(0,0,3)|}{\sqrt{6}} = \frac{\sqrt{45}}{\sqrt{6}} = \sqrt{\frac{15}{2}}$ units.

8)

The line is $\mathbf{r} = (0,0,3)+\lambda(2,1,-2)$. For $A = (6,0,4)$, the closest point satisfies

$$(2\lambda-6, \lambda, -1-2\lambda)\cdot(2,1,-2) = 0 \quad \Rightarrow \quad 9\lambda - 10 = 0, \quad \lambda = \frac{10}{9}.$$

The displacement from A to the line is $(-\frac{34}{9}, \frac{10}{9}, -\frac{29}{9})$, so the shortest distance is

$$\sqrt{\left(-\frac{34}{9}\right)^2 + \left(\frac{10}{9}\right)^2 + \left(-\frac{29}{9}\right)^2} = \frac{\sqrt{233}}{3} \text{ units.}$$

Alternatively, using the vector product,

$$h = \frac{|(2,1,-2)\times(6,0,1)|}{3} = \frac{|(1,-14,-6)|}{3} = \frac{\sqrt{233}}{3} \text{ units.}$$

9 a)

$\overrightarrow{AB} = (7, 13, 3)$, so the line through A and B has equation $\mathbf{r} = (2, -1, 1) + \lambda(7, 13, 3)$. Substituting the coordinates of P gives $\lambda = 1$ from the x-coordinate, $\lambda = -\frac{21}{13}$ from the y-coordinate, and $\lambda = -\frac{1}{3}$ from the z-coordinate. These are inconsistent, so P does not lie on l.

9 b)

Let $Q = (2 + 7\lambda, -1 + 13\lambda, 1 + 3\lambda)$. At the closest point, $\overrightarrow{PQ} \cdot (7, 13, 3) = 0$:

$$(-7 + 7\lambda, 21 + 13\lambda, 1 + 3\lambda) \cdot (7, 13, 3) = 0 \quad \Rightarrow \quad 227\lambda + 227 = 0, \quad \lambda = -1.$$

Thus $Q = (-5, -14, -2)$.

9 c)

$\overrightarrow{PQ} = (-14, 8, -2)$, and hence

$$PQ = \sqrt{(-14)^2 + 8^2 + (-2)^2} = 2\sqrt{66} \text{ units.}$$

Alternatively, for $\mathbf{d} = (7, 13, 3)$, $\mathbf{p} = (9, -22, 0)$ and $\mathbf{a} = (-2, -1, 1)$,

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{|(50, 28, -238)|}{\sqrt{227}} = 2\sqrt{66} \text{ units.}$$

10)

The midpoint of A and B is $(4, 0, 1)$, and $\overrightarrow{AB} = (2, 2, 2)$. The required line has a direction vector perpendicular to both $\overrightarrow{AB}$ and the plane normal $(2, 7, -9)$. Let its direction be (a, b, c). Then

$$2a + 7b - 9c = 0, \qquad a + b + c = 0.$$

Taking $c = 5$ gives $a = -16$ and $b = 11$. Therefore the line is

$$\mathbf{r} = (4, 0, 1) + \lambda(-16, 11, 5).$$

11 a)

Substitution gives

$$3(2) - 8(2) + 2(8) - 6 = 0, \qquad 3(10) - 8(3) + 2(0) - 6 = 0,$$

so both P and Q lie on Π.

11 b)

The midpoint of P and Q is $(6, \frac{5}{2}, 4)$, and $\overrightarrow{PQ} = (8, 1, -8)$. Let the required direction vector be (a, b, c), perpendicular to the plane normal $(3, -8, 2)$ and to $\overrightarrow{PQ}$. Thus

$$3a - 8b + 2c = 0, \qquad 8a + b - 8c = 0.$$

Taking $c = 67$ gives $a = 62$ and $b = 40$. The line is

$$\mathbf{r} = \left(6, \frac{5}{2}, 4\right) + \lambda(62, 40, 67).$$

11 c)

Write $R = (6 + 62\lambda, \frac{5}{2} + 40\lambda, 4 + 67\lambda)$. The condition $PR = \frac{43}{\sqrt{6}}$ gives

$$(4 + 62\lambda)^2 + (\tfrac{1}{2} + 40\lambda)^2 + (-4 + 67\lambda)^2 = \frac{1849}{6},$$

which simplifies to $9933\lambda^2 = \frac{3311}{12}$ and hence $\lambda = \pm\frac{1}{6}$. Substitution gives

$$R = \left(\frac{49}{3}, \frac{55}{6}, \frac{91}{6}\right) \quad \text{or} \quad R = \left(-\frac{13}{3}, -\frac{25}{6}, -\frac{43}{6}\right).$$

12 a)

$\overrightarrow{AB} = (1, 3, -1)$ and $\overrightarrow{AC} = (-2, 0, -1)$. A normal to the plane is

$$\overrightarrow{AB} \times \overrightarrow{AC} = (-3, 3, 6).$$

Thus the plane has equation $-3x + 3y + 6z = d$. Substituting $A(2, 1, 2)$ gives $d = 9$, so

$$-x + y + 2z = 3.$$

12 b)

The midpoint of A and B is $(\frac{5}{2}, \frac{5}{2}, \frac{3}{2})$ and $\overrightarrow{AB} = (1, 3, -1)$. A direction (a, b, c) for l_1 is perpendicular to both $(-1, 1, 2)$ and $\overrightarrow{AB}$:

$$-a + b + 2c = 0, \qquad a + 3b - c = 0.$$

Taking $c = 4$ gives $(a, b, c) = (7, -1, 4)$. Therefore

$$l_1 : \quad \mathbf{r} = \left(\frac{5}{2}, \frac{5}{2}, \frac{3}{2}\right) + \lambda(7, -1, 4).$$

12 c)

The midpoint of B and C is $(\frac{3}{2}, \frac{5}{2}, 1)$ and $\overrightarrow{BC} = (-3, -3, 0)$. A direction (a, b, c) for l_2 satisfies

$$-a + b + 2c = 0, \qquad a + b = 0.$$

Taking $c = 1$ gives $(a, b, c) = (1, -1, 1)$, so

$$l_2 : \quad \mathbf{r} = \left(\frac{3}{2}, \frac{5}{2}, 1\right) + \mu(1, -1, 1).$$

12 d)

Equating coordinates gives

$$7\lambda - \mu = -1, \qquad -\lambda + \mu = 0, \qquad 4\lambda - \mu = -\frac{1}{2}.$$

The first two equations give $\lambda = \mu = -\frac{1}{6}$, which also satisfies the third. The intersection is

$$P = \left(\frac{4}{3}, \frac{8}{3}, \frac{5}{6}\right).$$

13 a)

Substituting $Q = (10, k, 10)$ into the plane equation gives $10 - k + 60 = 300$, so $k = -230$.

13 b)

A point on the tunnel has position vector $(50 + 5\lambda, -765 + 5\lambda, -116 + \lambda)$. The vector from Q to this point is $(40 + 5\lambda, -535 + 5\lambda, -126 + \lambda)$. For the shortest distance this must be perpendicular to $(5, 5, 1)$:

$$(40 + 5\lambda, -535 + 5\lambda, -126 + \lambda) \cdot (5, 5, 1) = 0 \quad \Rightarrow \quad 51\lambda - 2601 = 0, \quad \lambda = 51.$$

Therefore $P = (305, -510, -65)$.

13 c)

$\overrightarrow{QP} = (295, -280, -75)$, so

$$QP = \sqrt{295^2 + (-280)^2 + (-75)^2} = 5\sqrt{6442} \text{ units.}$$

Alternatively, the vector-product method gives $h = |(5, 5, 1) \times (-40, 535, 126)|/\sqrt{51} = 5\sqrt{6442}$ units.

14 a)

A point on l has position vector $(1 + \lambda, 1 - \lambda, 2 + \lambda)$. For $P = (k, 2, -1)$ the displacement from P is $(1 - k + \lambda, -1 - \lambda, 3 + \lambda)$. Perpendicularity to $(1, -1, 1)$ gives $3\lambda - k + 5 = 0$, so $\lambda = (k - 5)/3$. The displacement at this value is $(-\frac{2k+2}{3}, \frac{2-k}{3}, \frac{k+4}{3})$. Hence

$$d = \frac{1}{3}\sqrt{(2k + 2)^2 + (2 - k)^2 + (k + 4)^2} = \frac{1}{3}\sqrt{6k^2 + 12k + 24}.$$

Alternatively,

$$d = \frac{|(1, -1, 1) \times (k - 1, 1, -3)|}{\sqrt{3}} = \frac{\sqrt{2k^2 + 4k + 8}}{\sqrt{3}} = \frac{1}{3}\sqrt{6k^2 + 12k + 24}.$$

14 b)

Completing the square,

$$d = \frac{1}{3}\sqrt{6(k + 1)^2 + 18},$$

which is minimised when $k = -1$, giving minimum distance $\sqrt{2}$.

15)

Let P be the foot of the perpendicular from A to the line. A point on the line is $(3\lambda, 1 + \lambda, \lambda)$, and

$$(3\lambda - 1, \lambda - 1, \lambda - 1) \cdot (3, 1, 1) = 0 \quad \Rightarrow \quad 11\lambda - 5 = 0, \quad \lambda = \frac{5}{11}.$$

Thus $\overrightarrow{AP} = (\frac{4}{11}, -\frac{6}{11}, -\frac{6}{11})$. The reflected image B satisfies $\overrightarrow{OB} = \overrightarrow{OA} + 2\overrightarrow{AP}$, so

$$B = \left(\frac{19}{11}, \frac{10}{11}, -\frac{1}{11}\right).$$

16)

A point on the line is $(1+2\lambda, 4\lambda, 3-3\lambda)$. The perpendicular from $A = (29, 29, -\frac{109}{3})$ meets the line when

$$(2\lambda - 28, 4\lambda - 29, \tfrac{118}{3} - 3\lambda) \cdot (2, 4, -3) = 0 \quad \Rightarrow \quad 29\lambda - 290 = 0, \quad \lambda = 10.$$

The foot is $(21, 40, -27)$ and $\overrightarrow{AP} = (-8, 11, \frac{28}{3})$. The reflected point is $B = A + 2\overrightarrow{AP}$, so

$$B = \left(13, 51, -\frac{53}{3}\right).$$

17)

Choose $A = (1, 0, 0)$ on l_1 and a point $P = (0, \mu, 1+\mu)$ on l_2. Perpendicularity of $\overrightarrow{AP} = (-1, \mu, 1+\mu)$ to the direction $(0, 1, 1)$ gives $2\mu + 1 = 0$, so $\mu = -\frac{1}{2}$. Therefore

$$|AP| = \sqrt{1 + \frac{1}{4} + \frac{1}{4}} = \frac{\sqrt{6}}{2} \text{ units.}$$

Alternatively,

$$h = \frac{|(0,1,1) \times (1,0,-1)|}{\sqrt{2}} = \frac{|(-1,1,-1)|}{\sqrt{2}} = \frac{\sqrt{6}}{2} \text{ units.}$$

18)

Choose $A = (1, 0, 1)$ on l_1 and $P = (1+\mu, -1+\mu, 1+2\mu)$ on l_2. Then $\overrightarrow{AP} = (\mu, -1+\mu, 2\mu)$ and

$$(\mu, -1+\mu, 2\mu) \cdot (1, 1, 2) = 0 \quad \Rightarrow \quad 6\mu - 1 = 0, \quad \mu = \frac{1}{6}.$$

Thus the distance is

$$|AP| = \sqrt{(\tfrac{1}{6})^2 + (-\tfrac{5}{6})^2 + (\tfrac{1}{3})^2} = \frac{\sqrt{30}}{6} \text{ units.}$$

Alternatively,

$$h = \frac{|(1,1,2) \times (0,1,0)|}{\sqrt{6}} = \frac{|(-2,0,1)|}{\sqrt{6}} = \frac{\sqrt{30}}{6} \text{ units.}$$

19)

The direction vectors are $\mathbf{d}_1 = (2, 1, 1)$ and $\mathbf{d}_2 = (1, 2, 2)$. Their cross product is $(0, -3, 3)$. With $\mathbf{a}_1 = (1, 0, 0)$ and $\mathbf{a}_2 = (0, 1, 0)$, the shortest distance is

$$h = \left| \frac{(0,-3,3)}{|(0,-3,3)|} \cdot (1, -1, 0) \right| = \frac{3}{\sqrt{18}} = \frac{\sqrt{2}}{2} \text{ units.}$$

20)

The direction vectors of the crane arms are $\mathbf{d}_1 = (60, 60, 30)$ and $\mathbf{d}_2 = (-30, 50, 30)$, with base points $\mathbf{a}_1 = (20, 20, 60)$ and $\mathbf{a}_2 = (60, 20, 70)$. Their cross product is $(300, -2700, 4800)$, and $\mathbf{a}_1 - \mathbf{a}_2 = (-40, 0, -10)$. Thus

$$h = \frac{|(300, -2700, 4800) \cdot (-40, 0, -10)|}{|(300, -2700, 4800)|} = \frac{60000}{3900\sqrt{2}} = \frac{100\sqrt{2}}{13} \text{ m.}$$

21)

Drone 1's path starts at $A_1 = (0, 9, 18)$ and has direction $(15, -3, -14)$. The first part of Drone 2's path starts at $B_1 = (15, 15, 3)$ and has direction $(-15, -15, 1)$. Their shortest distance is

$$h_1 = \frac{|((-213, 195, -270) \cdot (-15, -6, 15))|}{\sqrt{(-213)^2 + 195^2 + (-270)^2}} = \frac{2025}{\sqrt{156294}} = 5.12 \text{ m (3 SF)}.$$

For Drone 2's second segment, from $O_1 = (0, 0, 4)$ to $C_1 = (15, 12, 20)$, the direction is $(15, 12, 16)$. The cross product with Drone 1's direction is $(120, -450, 225)$, so

$$h_2 = \frac{|(120, -450, 225) \cdot (0, 9, 14)|}{\sqrt{120^2 + (-450)^2 + 225^2}} = \frac{900}{\sqrt{267525}} = 1.74 \text{ m (3 SF)}.$$

Therefore Drone 2's path is closer to Drone 1's path while travelling from above O to above C.

22 a)

The direction vectors of the ski lifts are $\mathbf{d}_1 = (40, 2, 1)$ and $\mathbf{d}_2 = (30, -2, 3)$, with base points $\mathbf{a}_1 = (-180, -20, 25)$ and $\mathbf{a}_2 = (-220, 44, -1)$. Their cross product is $(8, -90, -140)$ and $\mathbf{a}_1 - \mathbf{a}_2 = (40, -64, 26)$. Hence

$$h = \frac{|(8, -90, -140) \cdot (40, -64, 26)|}{\sqrt{8^2 + (-90)^2 + (-140)^2}} = \frac{2440}{\sqrt{27764}} = 14.6 \text{ m (3 SF)}.$$

As $14.6 > 10$, the ski lifts pass the safety regulation.

22 b)

The ski lifts have been treated as particles without dimensions, although they will likely be more than half of $4.643\ldots$ metres wide each, so the gap between them may be less than 10 metres. It has also been assumed that the ski lifts travel along straight-line paths.

23)

Take $\mathbf{a}_1 = (1, 0, 3)$, $\mathbf{d}_1 = (k, 2, 1)$, $\mathbf{a}_2 = (3, 2, 1)$ and $\mathbf{d}_2 = (4, 0, 1)$. Then

$$\mathbf{d}_1 \times \mathbf{d}_2 = (2, 4 - k, -8), \qquad \mathbf{a}_1 - \mathbf{a}_2 = (-2, -2, 2).$$

Using the shortest-distance formula for skew lines,

$$2 = \left| \frac{(2, 4 - k, -8) \cdot (-2, -2, 2)}{|(2, 4 - k, -8)|} \right| = \frac{|2k - 28|}{\sqrt{k^2 - 8k + 84}}.$$

Therefore

$$(2k - 28)^2 = 4(k^2 - 8k + 84) \quad \Rightarrow \quad 80k - 448 = 0,$$

so $k = \frac{28}{5}$.

24)

Let $\mathbf{a}_1 = (0, -1, 2)$, $\mathbf{d}_1 = (3, k, k)$, $\mathbf{a}_2 = (1, 0, 2)$ and $\mathbf{d}_2 = (1, -1, 0)$. Then

$$\mathbf{d}_1 \times \mathbf{d}_2 = (k, k, -3 - k), \qquad \mathbf{a}_1 - \mathbf{a}_2 = (-1, -1, 0).$$

Hence

$$1 = \frac{|(k, k, -3 - k) \cdot (-1, -1, 0)|}{\sqrt{k^2 + k^2 + (-3 - k)^2}} = \frac{|-2k|}{\sqrt{3k^2 + 6k + 9}}.$$

Squaring gives $k^2 - 6k - 9 = 0$, so $(k - 3)^2 = 18$ and $k = 3 \pm 3\sqrt{2}$.

25)

Let $\mathbf{a}_1 = (-1, k, 1)$, $\mathbf{d}_1 = (k, 2, -1)$, $\mathbf{a}_2 = (k, k, 1)$ and $\mathbf{d}_2 = (3k, -2, 2)$. Then

$$\mathbf{d}_1 \times \mathbf{d}_2 = (2, -5k, -8k), \qquad \mathbf{a}_1 - \mathbf{a}_2 = (-1 - k, 0, 0).$$

Thus

$$1 = \frac{|(2, -5k, -8k) \cdot (-1 - k, 0, 0)|}{\sqrt{4 + 25k^2 + 64k^2}} = \frac{|-2k - 2|}{\sqrt{89k^2 + 4}}.$$

Squaring gives $85k^2 - 8k = 0$, so $k = 0$ or $k = \frac{8}{85}$. Since k is positive, $k = \frac{8}{85}$.

26)

The direction vectors and base points are

$$\mathbf{a}_1 = (3.5, k, 2.5), \quad \mathbf{d}_1 = (-2, 0.5 - k, 0.5), \qquad \mathbf{a}_2 = (1.5, 1, 3), \quad \mathbf{d}_2 = (2.5, 0.5, -0.5).$$

Their cross product is

$$\mathbf{d}_1 \times \mathbf{d}_2 = \left(\frac{k}{2} - \frac{1}{2}, \frac{1}{4}, \frac{5k}{2} - \frac{9}{4}\right), \qquad \mathbf{a}_1 - \mathbf{a}_2 = \left(2, k - 1, -\frac{1}{2}\right).$$

The shortest distance is 0.4 m, so

$$\frac{\left|\left(\frac{k}{2} - \frac{1}{2}, \frac{1}{4}, \frac{5k}{2} - \frac{9}{4}\right) \cdot \left(2, k - 1, -\frac{1}{2}\right)\right|}{\sqrt{\left(\frac{k}{2} - \frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{5k}{2} - \frac{9}{4}\right)^2}} = 0.4.$$

The numerator simplifies to $\frac{1}{8}$ and the denominator squared to $\frac{13}{2}k^2 - \frac{47}{4}k + \frac{43}{8}$. Hence

$$\frac{1}{8} = \frac{2}{5}\sqrt{\frac{13}{2}k^2 - \frac{47}{4}k + \frac{43}{8}} \quad \Rightarrow \quad k^2 - \frac{47}{26}k + \frac{1351}{1664} = 0.$$

Completing the square gives $\left(k - \frac{47}{52}\right)^2 = \frac{109}{21632}$, and so

$$k = \frac{47}{52} \pm \frac{\sqrt{218}}{208}.$$

27)

Write the lines in vector form:

$$l_1 : \ \mathbf{r} = (-2, 5, -7) + \lambda(4, 2, 3), \qquad l_2 : \ \mathbf{r} = (1, -2, 4) + \mu(8, 4, 5).$$

The direction-vector cross product is

$$(4, 2, 3) \times (8, 4, 5) = (-2, 4, 0),$$

and the difference of the base points is $(-3, 7, -11)$. Therefore

$$h = \frac{|(-2, 4, 0) \cdot (-3, 7, -11)|}{\sqrt{(-2)^2 + 4^2}} = \frac{34}{\sqrt{20}} = \frac{17\sqrt{5}}{4} = 9.50 \text{ units (3 SF)}.$$

28)

Write the lines in vector form:

$$l_1:\ \mathbf{r} = (1,-3,0) + \lambda(1,5,2), \qquad l_2:\ \mathbf{r} = (-5,3,1) + \mu(2,1,-1).$$

Then

$$(1,5,2) \times (2,1,-1) = (-7,5,-9), \qquad (1,-3,0) - (-5,3,1) = (6,-6,-1).$$

Thus

$$h = \frac{|(-7,5,-9)\cdot(6,-6,-1)|}{\sqrt{(-7)^2+5^2+(-9)^2}} = \frac{63}{\sqrt{155}} = \frac{63\sqrt{155}}{155} \text{ units.}$$

29)

The lines can be written

$$l_1:\ \mathbf{r} = \left(-\frac{5}{2}, 4, \frac{5}{2}\right) + \lambda(1,-2,-1)$$
$$l_2:\ \mathbf{r} = (0,0,3) + \mu(8,1,0)$$

The direction-vector cross product is $(1,-8,17)$. The difference of the base points is $(-\frac{5}{2},4,-\frac{1}{2})$. Hence

$$h = \frac{|-42-30+9|}{\sqrt{155}} = \frac{63}{\sqrt{155}} = \frac{63\sqrt{155}}{155} \text{ units.}$$

30)

Write the lines as

$$l_1:\ \mathbf{r} = (1,2,4) + \lambda(1,-1,-1), \qquad l_2:\ \mathbf{r} = (1,0,0) + \mu(0,1,k).$$

Then

$$(1,-1,-1) \times (0,1,k) = (1-k,-k,1), \qquad (1,2,4) - (1,0,0) = (0,2,4).$$

Using the given distance,

$$\sqrt{2} = \frac{|(1-k,-k,1)\cdot(0,2,4)|}{\sqrt{(1-k)^2+k^2+1}} = \frac{|4-2k|}{\sqrt{2k^2-2k+2}}.$$

Squaring gives $4k^2-16k+16 = 4k^2-4k+4$, so $k=1$.

31 a)

The direction vectors of the two lines are $(-6,5,3)$ and $(6,-5,-3)$. Since one is the negative of the other, the lines are parallel.

31 b)

Take $A = (7,6,10)$ on l_1 and a point $P = (6\mu, 2-5\mu, -9-3\mu)$ on l_2. At the closest point, $\overrightarrow{AP}\cdot(6,-5,-3) = 0$:

$$(6\mu-7,-4-5\mu,-19-3\mu)\cdot(6,-5,-3) = 0 \quad \Rightarrow \quad 70\mu+35=0, \quad \mu = -\frac{1}{2}.$$

Thus $\overrightarrow{AP} = (-10,-\frac{3}{2},-\frac{35}{2})$, and

$$h = |AP| = \sqrt{(-10)^2+(-\tfrac{3}{2})^2+(-\tfrac{35}{2})^2} = \frac{\sqrt{1634}}{2} \text{ units.}$$

Alternatively, with $\mathbf{d} = (6,-5,-3)$ and the point $\mathbf{p} = (7,6,10)$ on the first line, and $\mathbf{a} = (0,2,-9)$ on the second,

$$h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{|(-83,-135,59)|}{\sqrt{70}} = \frac{\sqrt{1634}}{2} \text{ units.}$$

32)

The lines have base points $\mathbf{a}_1 = (2, 1, 0)$ and $\mathbf{a}_2 = (-1, 5, -3)$, and direction vectors $\mathbf{d}_1 = (0, -1, 3)$ and $\mathbf{d}_2 = (2, 5, -1)$. Then

$$\mathbf{d}_1 \times \mathbf{d}_2 = (-14, 6, 2), \qquad \mathbf{a}_1 - \mathbf{a}_2 = (3, -4, 3).$$

Therefore

$$h = \frac{|(-14, 6, 2) \cdot (3, -4, 3)|}{\sqrt{(-14)^2 + 6^2 + 2^2}} = \frac{60}{\sqrt{236}} = \frac{30\sqrt{59}}{59} \text{ units.}$$

33)

Write

$$l_1 : \ \mathbf{r} = \left(-\frac{3}{2}, 2, 3\right) + \lambda(2, 0, -1), \qquad l_2 : \ \mathbf{r} = (2, 0, 1) + \mu(-7, 2, 3).$$

The direction-vector cross product is $(2, 1, 4)$, and the difference of the base points is $(-\frac{7}{2}, 2, 2)$. Thus

$$h = \frac{|(2, 1, 4) \cdot (-\frac{7}{2}, 2, 2)|}{\sqrt{2^2 + 1^2 + 4^2}} = \frac{3}{\sqrt{21}} = \frac{\sqrt{21}}{7} \text{ units.}$$